

Publications by Jared Miller

I. JOURNAL PAPERS (PUBLISHED)

- 1) J. Miller, T. Dai, and M. Sznaier, “Robust Data-Driven Control of Discrete-Time Linear Systems with Errors in Variables,” *IEEE Transactions on Automatic Control*, pp. 1–15, 2024. [\[link\]](#) (Early Access)
- 2) J. Miller and R. S. Smith, “Peak estimation of rational systems using convex optimization,” *European Journal of Control*, p. 101088, 2024. [\[link\]](#) (2024 European Control Conference Special Issue)
- 3) J. Miller, T. Dai, and M. Sznaier, “Data-Driven Superstabilizing Control Under Quadratically-Bounded Errors-in-Variables Noise,” *IEEE Control Systems Letters*, vol. 8, pp. 1655–1660, 2024. [\[link\]](#)
- 4) J. Zheng, J. Miller, and M. Sznaier, “Data-Driven Safe Control of Discrete-Time Non-Linear Systems,” *IEEE Control Systems Letters*, vol. 8, pp. 1553–1558, 2024. [\[link\]](#)
- 5) J. Miller and M. Sznaier, “Bounding the Distance to Unsafe Sets with Convex Optimization,” *IEEE Transactions on Automatic Control*, vol. 68, no. 12, pp. 7575 – 7590, 2023. [\[link\]](#)
- 6) J. Zheng, T. Dai, J. Miller, and M. Sznaier, “Robust Data-Driven Safe Control using Density Functions,” *IEEE Control Systems Letters*, vol. 7, pp. 2611–2616, 2023. [\[link\]](#) (LCSS-CDC)
- 7) J. Miller and M. Sznaier, “Data-Driven Gain Scheduling Control of Linear Parameter-Varying Systems using Quadratic Matrix Inequalities,” *IEEE Control Systems Letters*, vol. 7, pp. 835–840, 2022. [\[link\]](#) (LCSS-ACC)
- 8) J. Miller, Y. Zheng, M. Sznaier, and A. Papachristodoulou, “Decomposed structured subsets for semidefinite and sum-of-squares optimization,” *Automatica*, vol. 137, pp. 110–125, 2022. [\[link\]](#)
- 9) J. Miller, D. Henrion, and M. Sznaier, “Peak Estimation Recovery and Safety Analysis,” *IEEE Control Systems Letters*, vol. 5, no. 6, pp. 1982–1987, 2021. [\[link\]](#) (LCSS-ACC)
- 10) J. Miller, M. A. Al-Radhawi, and E. D. Sontag, “Mediating Ribosomal Competition by Splitting Pools,” *IEEE Control Systems Letters*, vol. 5, no. 5, pp. 1555–1560, 2021. [\[link\]](#) (LCSS-ACC)

II. JOURNAL PAPERS (SUBMITTED)

- 1) J. Miller and M. Sznaier, “Quantifying the Safety of Trajectories using Peak-Minimizing Control,” 2023. [\[link\]](#)
- 2) J. Miller and M. Sznaier, “Analysis and Control of Input-Affine Dynamical Systems using Infinite-Dimensional Robust Counterparts,” 2023. [\[link\]](#)
- 3) J. Miller, M. Tacchi, M. Sznaier, and A. Jasour, “Peak Value-at-Risk Estimation of Stochastic Processes using Occupation Measures,” 2024. [\[link\]](#)
- 4) J. Miller, M. Tacchi, D. Henrion, and M. Sznaier, “Unsafe Probabilities and Risk Contours for Stochastic Processes using Convex Optimization,” 2024. [\[link\]](#)
- 5) J. Miller and M. Sznaier, “Peak Estimation of Hybrid Systems with Convex Optimization,” 2023. [\[link\]](#)

III. PHD THESIS

- 1) J. Miller, *Safety Quantification for Nonlinear and Time-Delay Systems using Occupation Measures*. PhD thesis, Northeastern University, April 2023. [\[link\]](#), [\[interview for Ph.D.s in Control\]](#)

IV. CONFERENCE PROCEEDINGS (PUBLISHED)

- 1) J. Miller, M. Korda, V. Magron, and M. Sznaier, “Peak Estimation of Time Delay Systems Using Occupation Measures,” in

2023 62nd IEEE Conference on Decision and Control (CDC), pp. 5294–5300, 2023. [\[link\]](#)

- 2) J. Miller, T. Dai, M. Sznaier, and B. Shafai, “Data-Driven Control of Positive Linear Systems using Linear Programming,” in 2023 62nd IEEE Conference on Decision and Control (CDC), pp. 1588–1594, 2023. [\[link\]](#)
- 3) J. Miller, M. Tacchi, M. Sznaier, and A. Jasour, “Peak Value-at-Risk Estimation for Stochastic Differential Equations Using Occupation Measures,” in 2023 62nd IEEE Conference on Decision and Control (CDC), pp. 4836–4842, 2023. [\[link\]](#)
- 4) J. Miller, T. Dai, and M. Sznaier, “Superstabilizing Control of Discrete-Time ARX Models under Error in Variables,” in 22nd IFAC World Congress, vol. 56, pp. 2444–2449, 2023. [\[link\]](#)
- 5) J. Miller, T. Dai, and M. Sznaier, “Data-Driven Superstabilizing Control of Error-in-Variables Discrete-Time Linear Systems,” in 2022 61st IEEE Conference on Decision and Control (CDC), pp. 4924–4929, 2022. [\[link\]](#)
- 6) J. Miller and M. Sznaier, “Bounding the Distance of Closest Approach to Unsafe Sets with Occupation Measures,” in 2022 61st IEEE Conference on Decision and Control (CDC), pp. 5008–5013, 2022. [\[link\]](#)
- 7) F. Bečanović, J. Miller, V. Bonnet, K. Jovanović, and S. Mohammed, “Assessing the Quality of a Set of Basis Functions for Inverse Optimal Control via Projection onto Global Minimizers,” in 2022 IEEE 61st Conference on Decision and Control (CDC), pp. 7598–7605, 2022. [\[link\]](#)
- 8) J. Miller and M. Sznaier, “Facial Input Decompositions for Robust Peak Estimation under Polyhedral Uncertainty,” *IFAC-PapersOnLine*, vol. 55, no. 25, pp. 55–60, 2022. [\[link\]](#)
- 9) J. Miller, D. Henrion, M. Sznaier, and M. Korda, “Peak Estimation for Uncertain and Switched Systems,” in 2021 60th IEEE Conference on Decision and Control (CDC), pp. 3222–3228, 2021. [\[link\]](#)
- 10) J. Miller, R. Singh, and M. Sznaier, “MIMO System Identification by Randomized Active-Set Methods,” in 2020 59th IEEE Conference on Decision and Control (CDC), pp. 2246–2251, 2020. [\[link\]](#)
- 11) J. Miller, Y. Zheng, M. Sznaier, and A. Papachristodoulou, “Decomposed Structured Subsets for Semidefinite Optimization,” in 2020 21st IFAC World Congress, 2020. [\[link\]](#)
- 12) C. Wu, J. Miller, Y. Chang, M. Sznaier, and J. Dy, “Solving Interpretable Kernel Dimensionality Reduction,” in *Advances in Neural Information Processing Systems* (H. Wallach, H. Larochelle, A. Beygelzimer, F. d’Alché-Buc, E. Fox, and R. Garnett, eds.), vol. 32, pp. 7915–7925, Curran Associates, Inc., 2019. [\[link\]](#)
- 13) J. Miller, Y. Zheng, B. Roig-Solvas, M. Sznaier, and A. Papachristodoulou, “Chordal Decomposition in Rank Minimized Semidefinite Programs with Applications to Subspace Clustering,” in 2019 IEEE 58th Conference on Decision and Control (CDC), pp. 4916–4921, 2019. [\[link\]](#)
- 14) J. Miller and B. Shafai, “A Model of Heave Dynamics for Bagged Air Cushioned Vehicles,” in 2019 IEEE Conference on Control Technology and Applications (CCTA), pp. 976–981, 2019. [\[link\]](#)
- 15) B. Taskazan, J. Miller, U. Inyang-Udoh, O. Camps, and M. Sznaier, “Domain Adaptation Based Fault Detection in Label Imbalanced Cyberphysical Systems,” in 2019 IEEE Conference on Control Technology and Applications (CCTA), pp. 142–147, 2019. [\[link\]](#)

V. CONFERENCE PROCEEDINGS (ACCEPTED)

- 1) J. Miller, N. Schmid, M. Tacchi, D. Henrion, and R. S. Smith,

“Peak Time-Windowed Mean Estimation using Convex Optimization,” 2024. Conference on Decision and Control

- 2) M. Abdalmoaty, J. Miller, M. Yin, and R. S. Smith, “Frequency-Domain Identification of Discrete-Time Systems using Sum-of-Rational Optimization,” in *IFAC Symposium on System Identification*, 2024. [\[link\]](#)
- 3) A. Honarpisheh, R. Singh, J. Miller, and M. Sznaier, “Low Order Systems in a Loewner Framework,” in *IFAC Symposium on System Identification*, 2024.
- 4) J. Miller, J. Zheng, M. Sznaier, and C. Hixenbaugh, “Data-Driven Superstabilization of Linear Systems under Quantization,” in *2024 American Control Conference (ACC)*, 2024. [\[link\]](#)

VI. PREPRINTS

- 1) J. Miller, N. Schmid, M. Tacchi, D. Henrion, and R. S. Smith, “Peak Time-Windowed Risk Estimation of Stochastic Processes,” 2024. [\[link\]](#)
- 2) J. Miller, C. Meroni, M. Tacchi, and M. Velasco, “Maximizing Slice-Volumes of Semialgebraic Sets using Sum-of-Squares Programming,” 2024. [\[link\]](#)
- 3) D. Henrion, J. Miller, and M. S. E. Din, “Algebraic Proofs of Path Disconnectedness using Time-Dependent Barrier Functions,” 2024. [\[link\]](#)
- 4) T. Imtiaz, M. Kohler, J. Miller, Z. Wang, M. Sznaier, O. Camps, and J. Dy, “SAIF: Sparse Adversarial and Interpretable Attack Framework,” 2022. [\[link\]](#)

VII. SEMINARS

- 1) “Frequency-Domain Identification of Discrete-Time Systems using Sum-of-Rational Optimization,” August 19-23, 2024, 26th International Symposium on Mathematical Theory of Networks and Systems, Cambridge, UK. [\[link\]](#)
- 2) “Analysis and Control of Input-Affine Systems using Parameterized Robust Counterparts,” August 19-23, 2024, 26th International Symposium on Mathematical Theory of Networks and Systems, Cambridge, UK. [\[link\]](#)
- 3) “Going beyond ODE systems with the moment-SOS hierarchy,” July 28-August 2, 2024, Polynomial Optimization for Nonlinear Dynamics: Theory, Algorithms, and Applications, Oberwolfach, DE. [\[link\]](#)
- 4) “Risk Analysis of Stochastic Processes using Occupation Measures,” April 11, 2024, H-CODE Seminar Series, Laboratory of Signals and Systems (L2S), Paris-Saclay, FR.
- 5) “Maximizing Slice-Volumes of Semialgebraic Sets using Sum-of-Squares Programming,” March 13, 2024, Moments and Polynomials: Applications and Theory Universitz of Konstanz, Konstanz, DE. [\[link\]](#)
- 6) “Data-Driven Safety Quantification using Robust Optimization,” February 16, 2024, Institute for Systems Theory and Automatic Control (IST), University of Stuttgart, Stuttgart, DE. [\[link\]](#)
- 7) “Data-Driven Safety Quantification using Robust Optimization,” October 18, 2023, Cybernetic Systems and Controls Lab (CSCL), Arizona State University, Tempe, AZ. [\[link\]](#)
- 8) “Risk analysis for stochastic processes using polynomial optimization,” October 15-18, 2023, Convex Relaxations for Polynomial Optimization, INFORMS Annual Meeting, Phoenix, AZ [\[link\]](#).
- 9) “Data-Driven Safety Quantification using Robust Optimization,” October 11, 2023, Safe Autonomy and Intelligent Distributed Systems (SAIDS) group, University of Southern California, Los Angeles, CA. [\[link\]](#)
- 10) “Data-Driven Safety Quantification using Infinite-Dimensional Robust Convex Optimization,” July 27, 2023, Konstanz Real Algebraic Geometry Seminar, University of Konstanz. [\[link\]](#)
- 11) “Data-Driven Safety Quantification using Infinite-Dimensional Robust Convex Optimization,” May 29, 2023, Student Seminar Series on Optimization, Control & Learning, UC San Diego. [\[link\]](#)

- 12) “Quantifying Safety under Uncertainty using Occupation Measures,” May 26, 2023, Control Seminars @ UCI, UC Irvine
- 13) “Data-Driven Safety Quantification using Infinite-Dimensional Robust Convex Optimization,” May 19, 2023, Multi-Robot Systems Lab Meeting, Stanford University. [\[link\]](#)
- 14) “Analysis and Control of Time-Delay Systems Using Polynomial Optimization,” May 14, 2023, MS14 Studying Dynamics using Polynomial Optimization Tools, SIAM Conference on Dynamical Systems. [\[link\]](#)
- 15) “Data-Driven Control under Input and Measurement Noise,” April 9, 2023, Oden Institute Seminar, UT Austin. [\[link\]](#)
- 16) “Safety Quantification for Nonlinear and Time-Delay Systems using Occupation Measures,” April 3, 2023, PhD Thesis Defense, Northeastern University. [\[link\]](#)
- 17) “Data-Driven Control under Input and Measurement Noise,” NYU MERIIT Lab Seminar Series, New York City, Feb 21, 2023. [\[link\]](#)
- 18) “Bounding the Distance to Unsafe Sets with Convex Optimization,” DCSD Rising Stars, 2nd Modeling, Estimation and Control Conference, Jersey City, October 2-5 2022. [\[link\]](#)
- 19) “Bounding distances to unsafe sets,” June 16, 2022, IfA Coffee Talks, ETH Zurich. [\[link\]](#)
- 20) “Bounding distances to unsafe sets,” June 14, 2022, LA3 Meeting, EPFL Lausanne. [\[link\]](#)
- 21) “Bounding distances to unsafe sets,” June 3, 2022, Journées SMAI MODE, University of Limoges (XLIM). [\[link\]](#)
- 22) Tutorials about Interior Point Methods, Polynomial Optimization, Frank-Wolfe algorithms and variations, and SDP approximations, May 16-20, Sparsity and Big Data in Control, Systems Identification, and Machine Learning, European Embedded Control Institute.
- 23) “Bounding distances to unsafe sets,” April 14, 2022, Conic Linear Optimization for Computer-Assisted Proofs, Mathematisches Forschungsinstitut Oberwolfach (MFO). [\[link\]](#)
- 24) “Bounding distances to unsafe sets,” June 28, 2021, Brainstorming days on measure and polynomial optimization (BrainPOP), LAAS-CNRS. [\[link\]](#)
- 25) “Data-Driven Peak and Reachability Set Estimation,” May 25, 2021, MS112 Methods of Learning Dynamical Systems for Control, SIAM Conference on Dynamical Systems. [\[link\]](#)
- 26) “Analysis and Control of Time-Delay Systems with Occupation Measures,” May 3, 2021, BrainPOP, LAAS-CNRS. [\[link\]](#)
- 27) “Exploiting Structure in Rank-Constrained and Approximated Semidefinite Programs,” December 19, 2019, TISEM Operations Research Seminar, Tilburg University. [\[link\]](#)

VIII. POSTER SESSIONS

- 1) “Risk Analysis of Stochastic Processes using Polynomial Optimization,” November 13, 2023, Future Trends in Polynomial Optimization, LAAS-CNRS, Toulouse, FR. [\[link\]](#)
- 2) “Frequency Domain Identification via Sum-of-Rational Optimization,” September 25, 2023, European Research Network System Identification (ERNSI), Stockholm, SE. [\[link\]](#)
- 3) “Safety Analysis and Control using Measures,” April 13, 2023, RISE 2023, Northeastern University. [\[link\]](#)
- 4) “Safety Analysis and Control using Measures,” February 27, 2023, PhD Research Expo, Northeastern University. [\[link\]](#)
- 5) “Diameter Constrained Minimum Spanning Graphs,” January 31, Current Themes of Discrete Optimization: Boot-camp for early-career researchers, ICERM. [\[link\]](#)
- 6) “Safety Analysis using Distance Estimation and Measures,” August 24, 2022, CLEVR-AI MURI Yearly Review Meeting, Northeastern University. [\[link\]](#)
- 7) “Exploiting SDP Structure Yields Tighter Approximations,” April 9, 2020, RISE, Northeastern University (remote). [\[link\]](#)
- 8) “Exploiting SDP Structure Yields Tighter Approximations,” February 24, 2020, IPAM Control, Learning and Optimization workshop, University of California, Los Angeles. [\[link\]](#)

- 9) “Chordal Decompositions in Rank Minimized SDPs.” May 30-31, 2019. Learning for Decision and Control (L4DC), Massachusetts Institute of Technology. [\[link\]](#)
- 10) “Chordal Decompositions in Rank Minimized SDPs.” May 10, 2019. New England Machine Learning Day, Northeastern University. [\[link\]](#)
- 11) “Scattered data interpolation through B-spline wavelets and the Elastic Net.” April 14, 2017. RISE, Northeastern University. [\[link\]](#)
- 12) “A parallelized Python-based Multi-Point Thomson Scattering analysis in NSTX-U.” October 29, 2014. 56th Annual APS Plasma Physics Conference, New Orleans. [\[link\]](#)

IX. POSTER SESSIONS (UPCOMING)

- 1) “Quantifying Trajectory Safety by the Minimum Data Corruption Needed to Crash.” September 30-October 2, 2024. Symposium on Systems Theory in Data and Optimization. Universität Stuttgart. [\[link\]](#) (extended abstract)